Geavanceerde computerarchitectuur
Lab sessions

(JLI1QS)

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1 Session 1 - Getting Acquainted with GPU Programming

In this lab session, our primary objective was to familiarize ourselves with GPU programming. Unlike CPU programming, which prioritizes sequential thread execution, GPU programming excels in executing multiple threads in parallel. Our focus was on understanding the structure of a program and how it can be optimized for GPU execution.
We began by examining how a program can be broken down into smaller sequential tasks, or threads. To expedite execution, these threads can be run in parallel on the GPU. Threads are organized into blocks, with each block being processed by a distinct GPU core. In this session, our task was to reverse the order of elements in an array.
Initially, we attempted this using only one block containing nine threads. With this setup, all threads had to execute sequentially, leading to nine execution cycles. Next, we divided the threads into three blocks, each containing three threads. This allowed three threads to execute simultaneously, reducing the total execution cycles to three. This can be seen in figure 1.
Finally, we further divided the threads into nine blocks, with one thread per block. This approach necessitated only a single execution cycle, resulting in significant time savings. Although the impact was minimal in this particular case, the advantage would become more pronounced with a substantial increase in the number of threads to be executed.

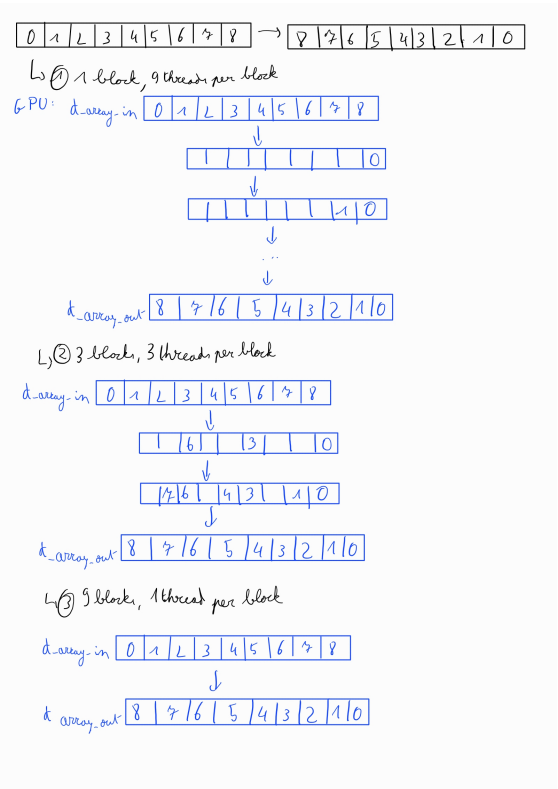

Figure 1: Illustration of blocks and threads

2 Session 2 - Race conditions

During the second lab session we had a look on race conditions and how to avoid them when GPU programming. We did a max detection of the elements of an array using three different methods. The race conditions in this assignment can occur when multiple threads attempt to access the array elements, but the threads of the previous step aren't finished yet. This can be illustrated by figure 3. When one of the threads in the first row hasn't finished yet and the threads of the second step would already start executing, this could give incorrect results. So after each row in figure 3 synchronization must be done to avoid this.
Fist we used a nested for loop on the CPU. For this method we used an extra variable where we stored the maximum value. We initialised this by setting this to the value of the first element of this array. Then using the for loop, we checked if each element of the array greater is then the maximum value variable. If so, we updated the maximum value to the value of the array element value. With this case no race conditions occur.
Secondly, we used a GPU kernel using the atomic max operation. An atomic operation gets executed without interruption.Threads get sequential access to a common resource. Because they have sequential access, there can be no interference and thus, as a result, no race conditions can occur. With the atomic max operation we had a time of 439 μs on an array of size 30000. just for comparison, the nested loop had and time of 2051 μs to find the maximum. So the atomic operations on a large array on the GPU are faster than a nested loop on a large array on the CPU.
Finally, we also used a GPU kernel using reduction. How we will exactly do this can be seen in figure 3, but instead of taking the sum of the elements, we take the maximum value of the two elements. More precisely, we loop over the array with a step size of 1 and search for the maximum value of two elements. The maximum value is placed in source 1 (see figure 2). Before we can continue and loop over the array with a greater step size, we have to synchronize threads first. Otherwise, some threads of the steps could not be finished yet and so we wouldn't get the correct the result. This is what we call race conditions and should be avoided. After the synchronisation we continue looping over the array, but each time with a greater step size as seen in figures 3 and 2. We stop looping when the step size is smaller then log_2(array_size).

Q: Is the result correct?
To fill the array, a loop going from 0 to aray size minus 1 was used each time. With nested loop and the atomic max, array of size 30 000 was used and a maximum was obtained of 29999. This is correct because the largest value contained in the array was 300000 - 1 = 29999.
With the GPU kernel, we had an array of size 2048. From this we got a maximum of 2047, this is correct as well!
Q: How does time increase as a function of input size?
With a very small array we saw that the CPU was always faster than the GPU, but with a huge array the GPU was much faster than the CPU. We can conclude from this that for small array operations the CPU is better but from that working with large arrays we better switch to the GPU. This can be declared by having a look at how CPU and GPU programming works, but this is already explained in section 1 (report of lab session 1).
Q: What happens if the input is larger than 2048 (how would you ensure correct calculation and are there additional synchronization steps)?
The first option is to use an atomic max operation to get the maximum of an array larger than 2048 elements. The atomic max operation is executed on the two maximum elements of each block (original array is splitted in two or more blocks). Secondly, we can also take the maximum of more then two elements within our kernel reduction method. This allows us to use larger

input arrays.

Q: Consider calling the kernel with much less threads than there are data elements. For example, one tenth. How would you ensure all elements get processed?

As mentioned in the answer above , we can split up the original array, so we can process a maximum amount of data elements with the given number of threads. After this first step we can reuse the kernel function to get the maximum element of all the maximum values which were stored in a new array.

Q: How would you tackle a 2D array?

For a 2D array we would take the maximum value of each column and then place them in a new 1D array. Tthis new array can then be used to find the max as in the previous examples for 1D-arrays. +

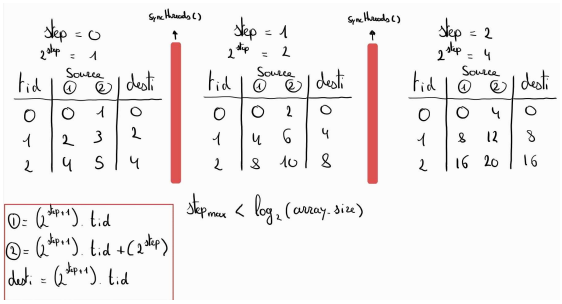

Figure 2: Illustration of reduction-method

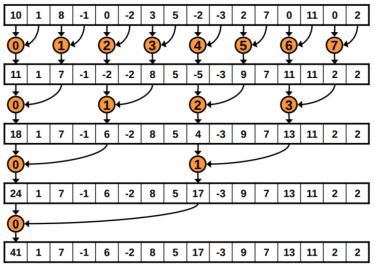

Figure 3: Illustration of array reduction

3 Session 3 - Warps, thread divergence, and striding

3.1 Warps

A warp is a group of 32 threads. These threads are grouped together for efficient execution of the same instruction simultaneously but operate on different data. So if we want to calculate the number of warps, we can do this with the following function: Number of warps = $(\frac{Threadsperblock}{32})\cdot$ number of blocks

In this first section, we are going to implement a kernel that inverts a image. To invert an image, we need to calculate the x = 255 - x for each color component.
When we use 1 thread, we become a execution time of 111854 μs (see figure 4). In this case, 111854 warps are created with a lot of GPU cores running idle. Each block gets a full warp assigned, but with only 1 thread per block the other 31 threads aren't used but still assigned for this block. When we are incrementing the amount of threads, then the execution time drops, because less warps are needed. When we are using 192 threads, we have a minimal excecution time of 71850 μseconds. This is the limit, because the GPU we used during this lab has only 192 cores and this is also the maximum of threads we can run in parallel.

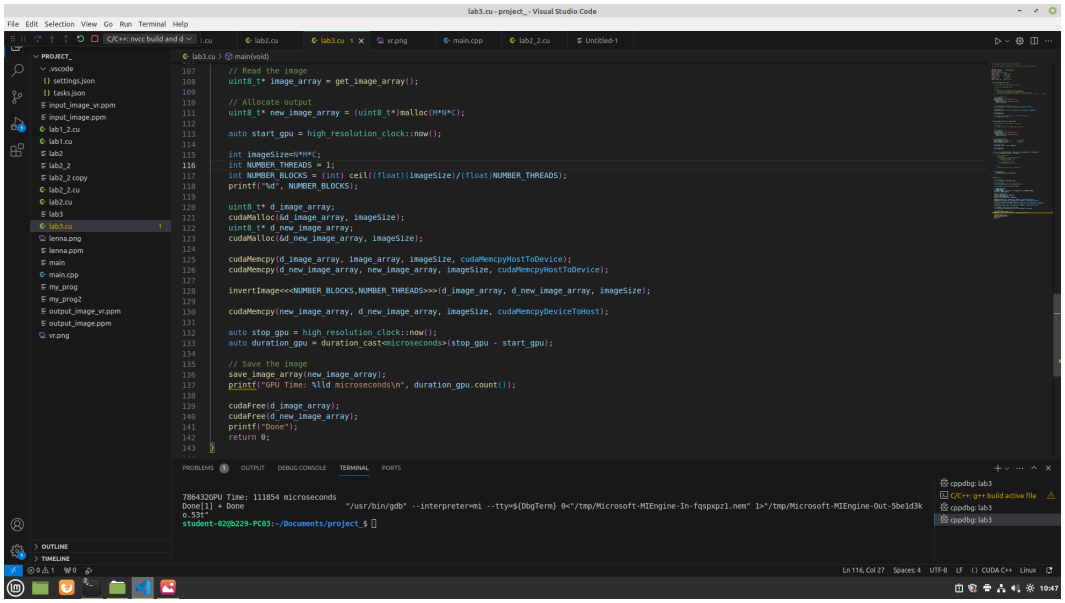

Figure 4: Excecution time 1 thread

3.2 Thread divergence

One part of the warp may have higher complexity and thus longer computation time. Because the other parts of the warp become free, they temporarily no longer contribute to the processing of the data but still possess a core.... We can solve this by first starting to split the incoming array in the code by R, G and B values. As seen in the task, the computational cost is higher for the R values than for the G and B values. Therefore, we are actually going to create a new

second kernel in order to compute the values separately so that no thread divergence occurs and the GPU cores are used optimally.

3.3 Some pointers - nonobligatory

3.3.1 Implement the inverse kernel and use the provided image load/save function to invert an image. You can use as many threads as there are input elements.

Figure 5: Image lenna

Figure 6: Inversed image lenna

3.3.2 Implementation of striding in the kernel

Striding occurs when the matching between threads and elements is not 1 to 1. This can cause the need to compute multiple elements in one thread. We can do this by checking which iteration we are in (starting from 0), multiplying this by the size of the grid of threads and then adding this to the initial index of the element. We do this until the new index is 1 less than the length of the incoming array. This was then implemented on an array with 16 elements and 12 threads. (See figure 7.) This involved adding 1 to each element using only 12 threads for all the 16 elements and obtaining the correct result.

In figure 8, we can see that the execution time is decreasing with the amount of threads. So we can conclude that if we use more threads, the execution will happen faster then if we use just a few threads.

Figure 7: Example striding

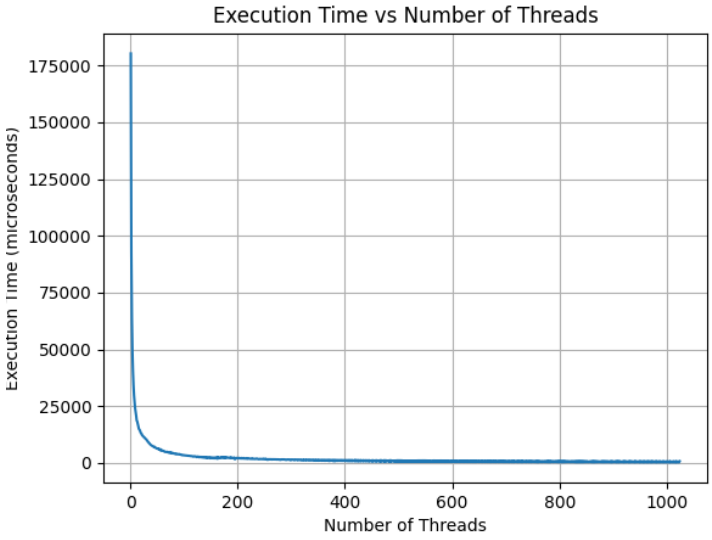

Figure 8: Execution time vs number of threads

3.3.3 Second kernel where the R component is processed according to $x = (x\%25) \cdot 10, x \in (100, 200)$

Figure 9: image lenna

Figure 10: second inversed image lenna

After this we made a third kernel, which instead of processing RGBRGB...RGB operates on an input format of RR...RGGG...GBB...B.
If we inverse the image with the additional conditions, we had a time of 76435μs. Now with the reformation of the input array we compute the R seperatly of the G and B (who need less computation). Each warp consist now of threads with equal computation costs, which reduces the GPU cores running idle. In total, this can give us a significant time gain.

4 Session 4 - Memory management

4.1 Part 1 - Coalesced memory access

As mentioned in earlier parts, threads are grouped in warps. Threads in the same warp mostly access adjacent memory addresses. Coalesced memory access allows us to combine multiple memory accesses into a single memory transaction. This reducess the number of required transactions and improves the execution time of a GPU program.

Figure 11: image lenna

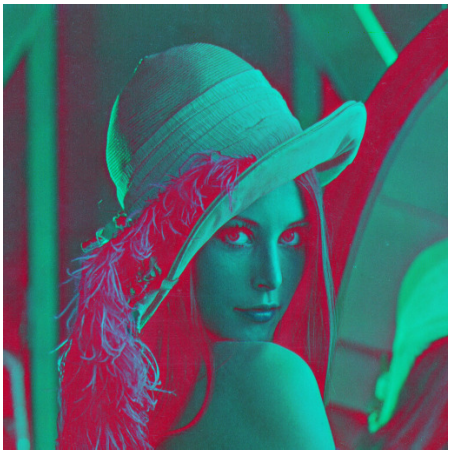

Figure 12: R-value inversed on lenna

As seen in figure 13, coalesced memory acces speeds up the execution in comparison with un-coalesced memory access. Even for smaller grid-sizes we can see an advantage by using coalesced memory access.

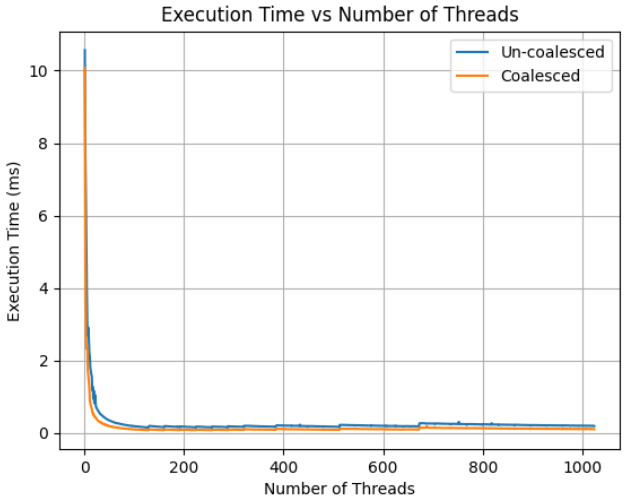

Figure 13: execution time un-coalesced vs coalesced

4.2 Part 2 – Global, shared, and constant memory

As second part of this lab session, we did matrix multiplications using different types of memory. Firstly, we used global memory to store our matrices. This memory is large and can be accessed by all functions, but is slow in comparison with other types. Secondly, we stored our matrices in shared memory. Shared memory is a small but very fast type of memory that is shared among threads within the same block. As third part, we also tried storin a matrix in constant memory. This type of memory is located off-device and optimized for low-latency read operations. Constant memory is typically used to store data that is constant throughout the execution of a kernel. More concrete, we started by using 2x2 matrices and in figure 14 the input matrices A and B are visible. Matrix multiplication is a mathematical operation where the elements of rows in one matrix are multiplied with corresponding elements of columns in another matrix, and the results are summed to generate a new matrix. So in our case, the matrix multiplication can be done as following: 3*3 + 6*6 = 45, 3*5 + 6*2 = 27, 7*3 + 5*6 = 51, 7*5 + 5*2 = 45. These values are equal to those we get in our program, so we can conclude our matrix multiplication is done correctly.

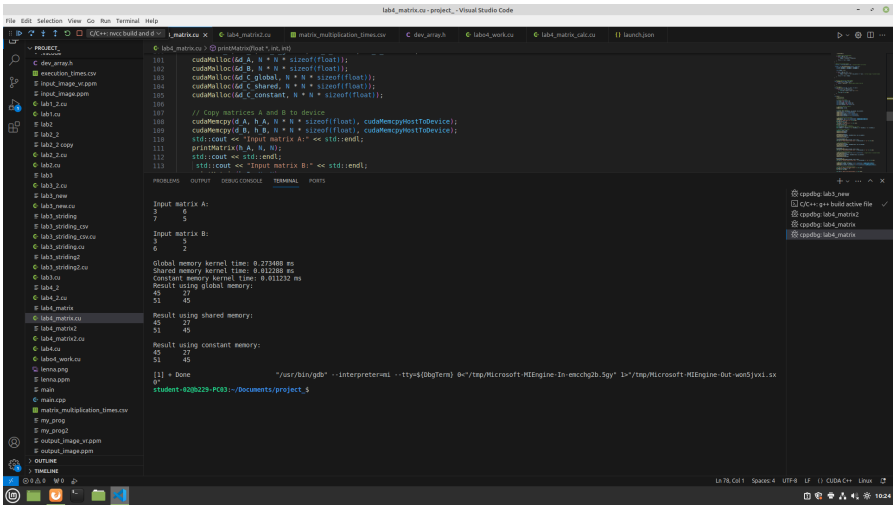

Figure 14: Verification matrix multiplication

By increasing the size of the matrices, we observe constant memory is significantly slower then shared and global memory. The results can be found in figure 15. When we want to continue using larger matrices, we need to split up the original matrix in smaller matrices of 32x32 elements, because of the limited size of the shared memory (mostly 16-48kbytes). As visible in figure 16, shared memory becomes faster and faster when going to a matrix size of 256x256 elements.

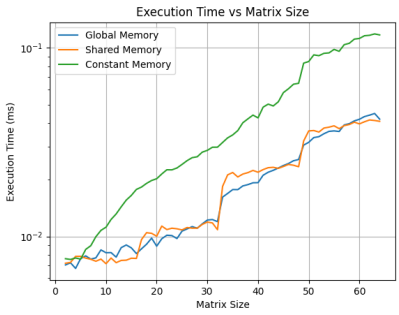

Figure 15: Execution time matrix size until 64x64

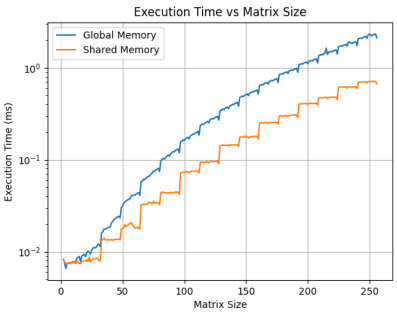

Figure 16: Execution time shared vs. global

5 Session 5 - Dynamic parallelism and asynchronous operation

5.1 Dynamic parallelism

Dynamic parallelism refers to the capability of GPUs to create and manage parallel tasks or threads on-the-fly within a kernel. For example, the GPU can spawn a kernel from within a kernel, enabling more flexible and complex program structures. This isn't executed during the lab sessions, because this requires at least compute capability 3.5, but the PCs in the lab only have 3.0.

5.2 Asynchronous operation

In GPU programming, asynchronous operation means that the GPU can execute independently at the same time of the CPU. Using this principle, we can maximize the CPU & GPU utilization without waiting for each other. This can reduce the execution time of a program. In the lab, we used 4 input arrays and 4 kernel functions to find the sum, product, minimum and maximum of these arrays. Firstly, we executed the full program synchronously: create an array (CPU), execute kernel function (GPU), create an array (CPU) ... Before executing the next step, we wait for the current step to finish. Secondly, we also tried doing this asynchronously. In stead of waiting for the GPU to finish, we already start creating the next array on the CPU. The total execution time for different array sizes are visible in figure 17.

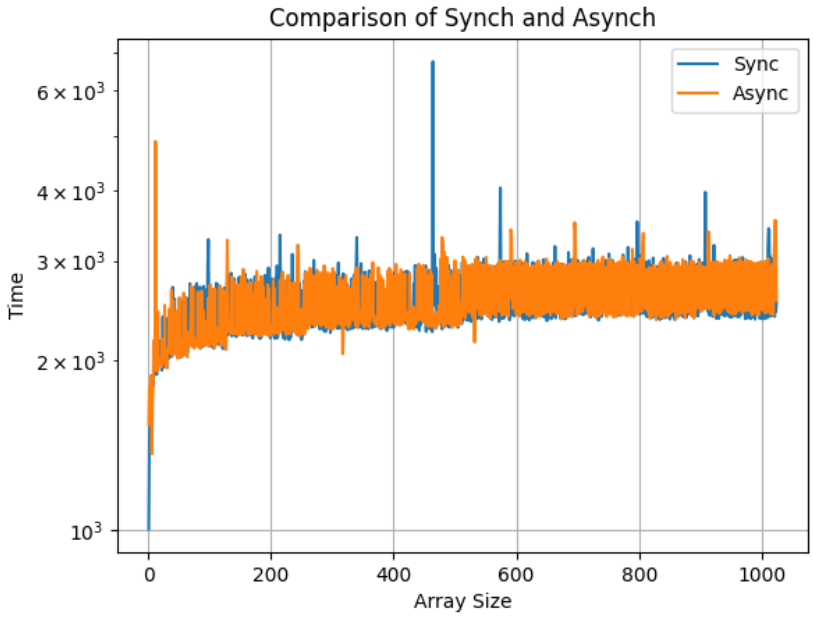

Figure 17: Synchronous vs Asynchronous Operation

Figure 17 doesn't give us a proper conclusion as both graphs are almost equivalent. To understand why the graphs are similar, we measured the execution time of the CPU and GPU

separately. In figure 18 we observe that the GPU execution time is much larger than the CPU time. Using asynchronous operation will not give us a big advantage, because the CPU execution time can almost be neglected. The CPU times are so small, that we could say that the CPU time can be equal to a measurement error on the total execution time.

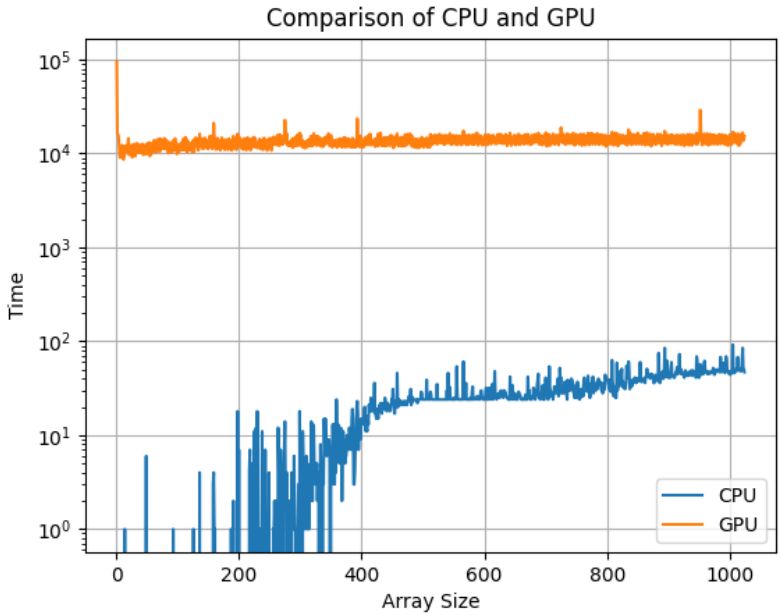

Figure 18: CPU time vs GPU time

Based on figure 18, it's not immediately evident that asynchronous operations provide a significant time advantage. This is due to our system operating very briefly on the CPU, resulting in only a marginal time benefit. However, it becomes apparent that if we were to execute a longer operation on the CPU and then proceed asynchronously, we could indeed gain a noticeable time advantage.

6 Project: Image stitching

6.1 Brainstorm

During our brainstormsessions we thought about a project where GPU programming had a lot of advantages...Both of us wanted to do something with our master thesis at first, but because we both work with small amounts of captured data in our IoT-system and no advanced data processing is required, we shelved this idea.
We started looking at other ideas. Both of us had interest in combining 2 pictures whith an overlap as is done when taking a panorama picture with your smartphone. If we then look further, we also immediately find that we can actually start running this in parallel on the GPU and so, for example, calculate row by row which allows us to use the GPU to its fullest.

6.2 Implementation

Starting with this project, we set some stages to become a fully usable application at the end. The first step started by loading a picture as was done in lab 2. This method of loading an image gives us one long array with RGB values. Since the 'Lenna' photo from the lab sessions has a width of 512 pixels and a height of 512 pixels, we obtain an array containing 786432 elements.
As second step we started by checking how we could merge 2 arrays. To find the overlap in the picture, we started by initiliazing 2 arrays with some equal data. By finding the correlation between the first values of the second array with two elements of the first array, we could examine the index to stitch.
As next step we started by extracting two 2D-arrays who represent the two images we want to stitch. The first image contains the first 300 pixels of each row of the loaded image and the second array contains the last 300 pixels of each row. The concept is shown visually in figure 19.

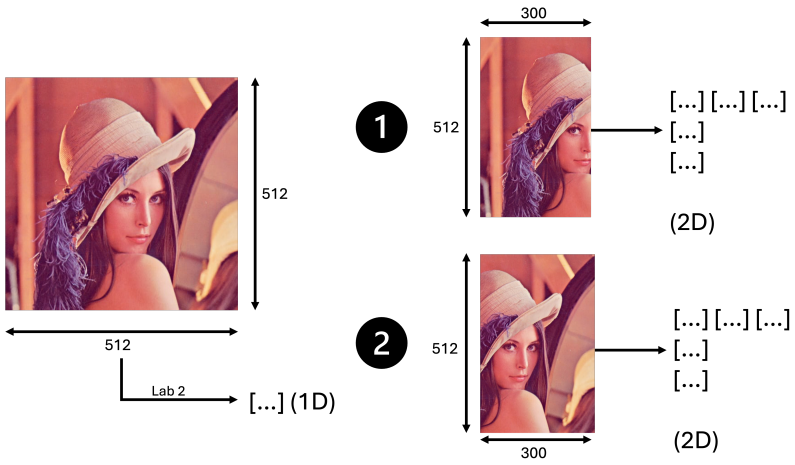

Figure 19: Concept first implementation

After splitting the loaded image into two 2D arrays, we can now proceed to the next step where we want to find the index to stitch the two images. In this step, we start by working row by row.
At first we wanted to calculate the correlation between pixels from both image, but as this process would be complex to implement, we started by thinking about other methods. By calculating the difference between pixels, comparing each pixel of the first image to the first pixel of the second image. This comparison helps determine the optimal alignment point for stitching the images together. This principle is shown in figure 20. To ensure that the thread index is within the bounds of the array and aligned with the RGB channel structure, a conditional check `if (idx < size - 5 && idx % 3 == 0)` is used. This condition verifies that there are enough elements for the comparisons and that the index aligns with the three-channel (R, G, B) structure of the image. The kernel then calculates the differences between the RGB values of the current two pixels in `arr1` and the first two pixels in `arr2`. The differences for each color channel are calculated separately with the following formulas:

$$\text{int diff_R} = |\text{arr1[idx]} - \text{arr2[0]}| + |\text{arr1[idx+3]} - \text{arr2[3]}|$$

$$\text{int diff_G} = |\text{arr1[idx + 1]} - \text{arr2[1]}| + |\text{arr1[idx + 4]} - \text{arr2[4]}|$$

$$\text{int diff_B} = |\text{arr1[idx + 2]} - \text{arr2[2]}| + |\text{arr1[idx + 5]} - \text{arr2[5]}|$$

These differences are summed up to obtain the total difference for the current pixels. The total difference is then stored in the `diffs` array, with each entry corresponding to a pixel.After computing the differences, a synchronization call `__syncthreads()` is used to ensure that all threads within a block complete their computations before proceeding.

Figure 20: calculation of the difference between corresponding pixels

Once the differences have been calculated, the next step is to find the minimum value in the differences array. This task is performed by the `get_minimum` CUDA kernel. The kernel iterates through the differences array and identifies the smallest value using an atomic operation to ensure that the minimum value is correctly determined even when multiple threads are involved. The result is the minimum difference value, which indicates the pixel in the first image that has the smallest color difference from the first two pixels of the second image.
After determining the minimum difference value, the `get_index` CUDA kernel is used to find the index of this minimum value in the differences array. This index represents the best stitching point, as it corresponds to the pixels in the first image that aligns most closely with the first two pixels of the second image in terms of color similarity.
After this, the most frequent stitching point across all rows is identified to ensure consistency and optimal alignment across the entire image. This is achieved by iterating through the array

of minimum indices for each row and finding the most frequently occurring index. This index is then used as the stitching point for the images.

The program initializes an array to hold the pixel data of the stitched image. The width of this new image is calculated as the sum of the width of the second image and the optimal stitching point index. Then it iterates through each row of the images to populate the stitched image array. For each row, it copies the pixel data from the first image up to the stitching point. After reaching the stitching point, the program continues by copying the pixel data from the second image, starting from the first column of the second image. Specifically, for each pixel, the program checks whether the current column index is less than the stitching point index. If it is, the pixel data from the first image is copied to the stitched image array. Otherwise, the pixel data from the second image is copied, adjusted by subtracting the stitching point index from the current column index to ensure proper alignment.
After constructing the stitched image array, the program saves the resulting image in PPM-format. Now that there is a working program, several images were used to go over where the program already works and where improvements are needed. The outputs of the images can be found in section 6.4.

6.3 Future work

Currently we are able to stitch images with our own program. If we look at the figure 21 and 22, we see that the program perfectly stitches the 2 images together. If we then look at the figure 23, we immediately see that if we take a picture ourselves, we get a very strange output. If we look at figure 24 we do have a good output. Only you can clearly see that this one is stitched together. If we want to stitch pictures that we took ourself as well, instead of starting to compare pixels at row level, we need to start comparing with surrounding rows. This would allow us to still stitch photos that are not perfectly lined up in the right place (vertical offset). Some interpolation between surrounding pixels values could possibly also give better results.

6.4 Examples

6.4.1 Example 1

First, we used the photo of Lenna , which was also used in the labs. Here we cut the image into 2 parts with some part overlapping. Finally we obtained back the original photo

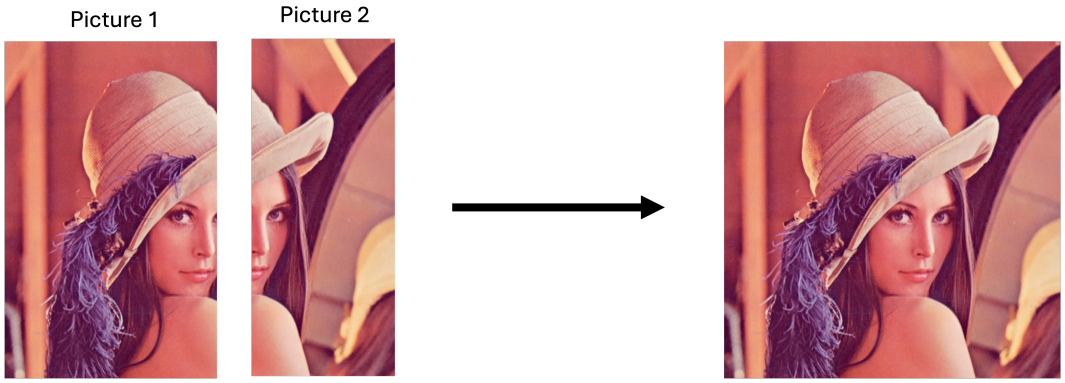

Figure 21: Example 1

6.4.2 Example 2

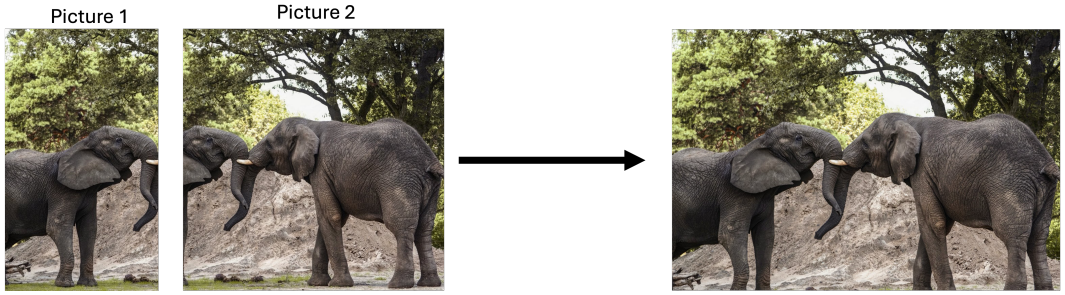

Figure 22: Example 2

6.4.3 Example 3

In the 3rd example, we took a picture of a multimeter where there is overlap. When we plugged it into the program, we got a discoloration as output. We suspect that there is another error somewhere in the code that does not return the R value of the RGB correctly. However, this is not the case with the other pictures...

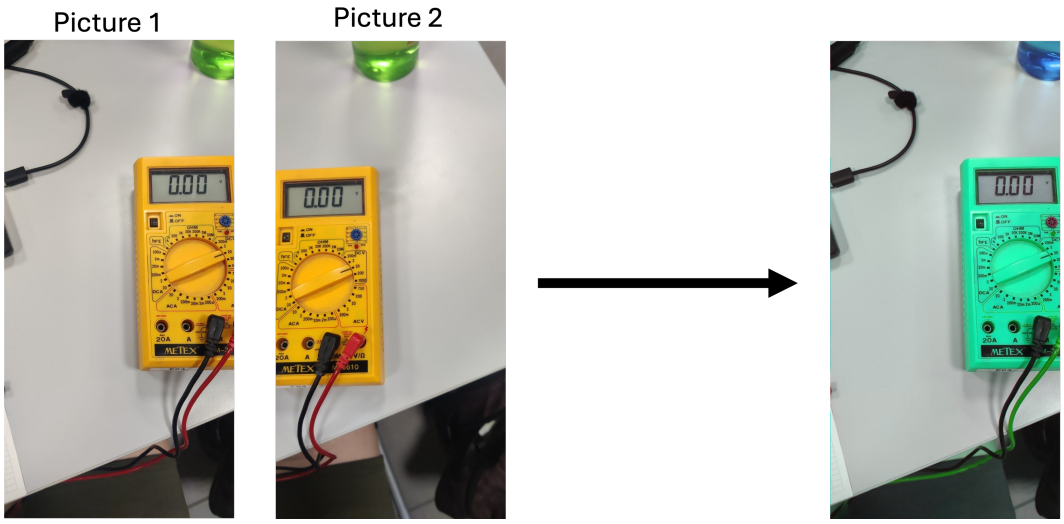

Figure 23: Example 3

6.4.4 Example 4

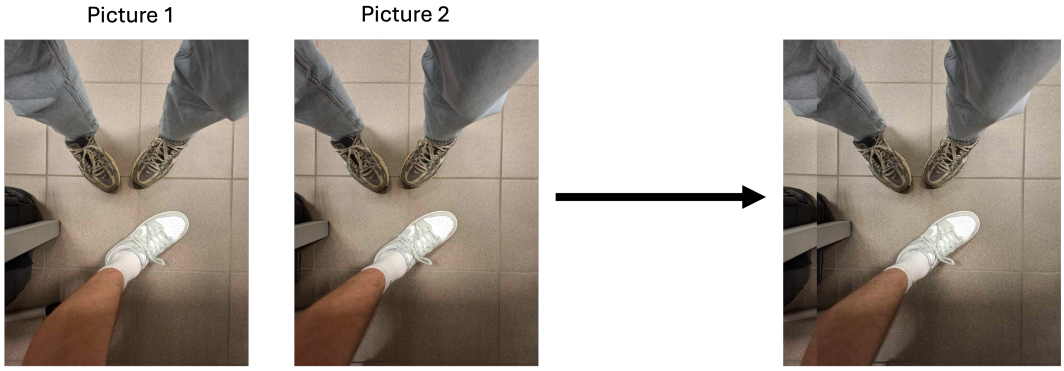

Figure 24: Example 4